

**FMCW Radar Shape Classification via Convolutional
Neural Networks:
A Simulation-Based Machine Learning Approach**

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Abstract

Frequency-Modulated Continuous-Wave (FMCW) radar offers a privacy-preserving, lighting-independent alternative to cameras for object recognition. However, the ambiguity of radar reflections makes classifying real-world objects challenging. This study narrows the scope to foundational 2D geometric shapes (Circle, Square, Rectangle, Triangle, and Oval) to establish a baseline for radar-based object classification. A physics-based simulation pipeline produced Range-Angle heatmaps from a simulated 77 GHz radar, with domain randomization across orientation, size, range, lateral position, and signal-to-noise ratio to support sim-to-real transferability. To assess the trade-off between hardware cost and classification accuracy, datasets were generated across multiple receive antenna configurations and classified using a 2D Convolutional Neural Network. Counter to conventional radar intuition, results revealed that higher angular resolution caused classification accuracy to decline at higher antenna counts. This report details the methodology, empirical results, and challenges of scaling machine learning models alongside advancing radar hardware.

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1. Introduction

1.1 Background and Motivation

Recently, radar systems have become widely accessible to researchers and everyday consumers due to their low cost and high-performance output. One application domain that has benefited most from this development is smart-home monitoring for elderly care. This is an area in which the privacy-preserving nature of radar offers an advantage over camera-based alternatives.

Unlike optical sensors, an FMCW radar produces no identifiable image of a person, operates in complete darkness, penetrates light obstructions and remains unaffected by ambient lighting conditions.

Despite these advantages, the practical adoption of radar for indoor object recognition has been hindered by a fundamental data problem. Raw radar returns are ambiguous. A one-dimensional time-domain Intermediate Frequency (IF) signal contains, in principle, information about an object's range, velocity, angular position, size, material, and shape, but extracting any of these properties requires a sequence of Fourier transforms followed by complex downstream interpretation. For static objects, this interpretation problem becomes even worse because the Doppler dimension carries no discriminative information. The major obstacle of applying machine learning to radar data for object recognition is the scarcity and time-consuming nature of collecting and labelling real-world radar datasets.

1.2 Problem Statement

This study addressed the following research question: Given only an FMCW Range-Angle heatmap, can a Convolutional Neural Network reliably classify the underlying geometric shape

of a physical object, and how does classification performance scale with the number of receive antennas in the radar array?

The original objective of the broader research is for general object recognition, in which a model must distinguish different configurations of an object in a noisy environment. However, objects are complex with poorly characterized radar cross-section, and training a classifier directly is intractable without a large real-world dataset. Geometric shapes provided an ideal abstraction layer between raw radar physics and real-world object classification. Every real object has a dominant geometric cross-section; a car is roughly a rectangle, a ball is a circle, etc. By teaching a model to classify shapes first, we establish a fundamental reasoning that should transfer directly to real-object recognition once a sim-to-real validation pipeline is established.

A second contribution of this work concerns hardware design. FMCW radar accuracy is widely believed to scale with the number of receive antennas because more antennas produce finer angular resolution. Yet the cost of radar systems (both monetary and in terms of system complexity) scales linearly with the number of antennas. A clear empirical answer to the question "how many antennas are needed for reliable object recognition?" would help consumers decide how much hardware they actually need for their use case.

1.3 Hypothesis and Project Scope

The working hypothesis at the beginning of this study held that classification accuracy would increase with an increase in the number of receive antennas, because higher antenna counts produce finer angular resolution and therefore more discriminative Range-Angle heatmaps. The empirical results contradicted this hypothesis in a manner that is both surprising and physically explainable.

The scope of this study was deliberately constrained to ensure clean experimental isolation. Five geometric shape classes were considered (Circle, Square, Rectangle, Triangle, Oval). All objects were treated as static and composed of the same material (metal). All training data was generated through a physics-based MATLAB simulation, with preliminary sim-to-real validation conducted using the MATLAB Radar Toolbox and Tinkercad-derived STL meshes. Real-hardware experiments using a Texas Instruments IWR1642 evaluation board were planned as future work and are discussed later.

2. Literature Review

2.1 FMCW Radar Signal Processing

The theoretical foundation for this work was derived from the Intro to Wireless Communication Systems (ECE/COS 368) lecture notes prepared by Professor Ghasempour^[1]. An FMCW radar transmits a chirp signal: a sinusoid whose frequency increases linearly with time over a bandwidth B during a chirp duration T_c , producing a slope $S = B / T_c$. When the chirp reflects from a target at range d , the round-trip delay $\tau = 2d/c$ maps to a constant beat frequency in the received IF signal, given by $f_{\text{beat}} = S \cdot 2d / c$. A Range Fast Fourier Transform (FFT) applied across the fast-time samples of a single chirp therefore produces a frequency-domain representation in which peaks correspond to target ranges. The range resolution of the resulting profile is given by $d_{\text{res}} = c / (2B)$, and the maximum unambiguous range by $d_{\text{max}} = (F_s \cdot c) / (2S)$. For the parameters used in this study ($B = 4 \text{ GHz}$, $T_c = 40 \text{ }\mu\text{s}$, $F_s = 4 \text{ MHz}$), these formulae yielded $d_{\text{res}} = 3.75 \text{ cm}$ and $d_{\text{max}} = 6.0 \text{ m}$, both of which are consistent with the operating envelope of indoor objects.

Velocity estimation in FMCW radar relies on phase coherence across multiple consecutive chirps. The phase change $\Delta\phi$ between consecutive chirps is related to target radial velocity V by $V = \lambda \Delta\phi / (4\pi T_c)$, and a Doppler FFT applied across slow time produces a Range-Doppler map. For static-object classification, however, the Doppler dimension provides no discriminative signal, and the analysis collapses to a Range-Angle representation.

Angular estimation requires a multi-element receive array. When a planar wavefront arrives at angle θ relative to the array, neighboring antennas observe a phase difference $\Delta\omega = (2\pi D \sin \theta) / \lambda$, where D is the antenna spacing. An Angle FFT applied across the antenna dimension recovers the angular spectrum of incident energy. The maximum field of view is achieved with

half-wavelength antenna spacing ($D = \lambda / 2$), which provides $\pm 90^\circ$ coverage. The angular resolution is governed by $\theta_{\text{res}} = \lambda / (N D \cos \theta)$, which is inversely proportional to the number of antennas N . For half-wavelength spacing, this simplifies to approximately $\theta_{\text{res}} \approx 114.6^\circ / N$, which yielded the values 28.6° , 14.3° , 11.5° , 9.6° , 7.2° , and 3.6° for the antenna counts evaluated in this study.

2.2 Geometric Scattering Models

The decision to model shapes as collections of point scatterers placed along edges and corners draws on electromagnetic scattering theory. For radar wavelengths much smaller than object dimensions, reflection occurs predominantly at boundaries rather than through the interior of an object. Corner reflectors are among the strongest known passive radar targets. Flat metal edges and curved surfaces reflect progressively less energy. This physical understanding motivated the amplitude assignments used in the scatterer models.

A central gap that this study addressed concerns the study of antenna-count effects on radar-based classification accuracy. While prior work has explored the relationship between antenna count and angular resolution from a signal processing standpoint, the question of how learning-based classifiers respond to increasing antenna count, and in particular, whether more antennas always produce better classification accuracy has not been widely explored. This study fills that gap by presenting a controlled comparison of CNN classifiers trained on identical shape datasets generated with six different antenna counts.

3. Methodology

3.1 Radar System Parameters

The simulated FMCW radar system used parameters consistent with the Texas Instruments IWR1642 evaluation board^[2], a 77 GHz single-chip FMCW radar widely used in academic research and commercial sensing applications. The carrier frequency was set to $f_c = 77$ GHz, the chirp bandwidth to $B = 4$ GHz, and the chirp duration to $T_c = 40$ μ s, yielding a slope of $S = 100$ GHz/s. The Analog-to-Digital Converter (ADC) sample rate was set to $F_s = 4$ MHz, producing 160 samples per chirp. Antenna spacing was fixed at $D = \lambda/2 \approx 1.95$ mm to maximize the field of view. The number of receive antennas N_{ant} was treated as the primary independent variable, taking values $\{4, 8, 10, 12, 16, 32\}$ across six parallel dataset generations. Table 1 summarizes the resulting derived performance metrics.

Table 1

Parameter	Symbol	Value
Carrier frequency	f_c	77 GHz
Bandwidth	B	4 GHz
Chirp duration	T_c	40 μ s
Slope	$S = B / T_c$	100 GHz/s
ADC sample rate	F_s	4 MHz
Range resolution	$d_{\text{res}} = c/2B$	3.75 cm
Maximum range	d_{max}	6.0 m
Antenna spacing	D	$\lambda/2 \approx 1.95$ mm
Receive antenna counts	N_{ant}	$\{4, 8, 10, 12, 16, 32\}$

3.2 Geometric Scatterer Models

Each shape was modeled as a collection of point scatterers, each represented by a tuple (x, y, amplitude) specifying the scatterer's spatial position relative to the object's center and its reflectivity. Amplitude assignments followed physical principles described earlier. Corner reflectors received amplitude 1.0 (the strongest), flat edges received amplitude 0.6 (intermediate), and curved surfaces received amplitudes between 0.3 and 0.5. The five shape models were constructed as follows:

- Circle was modeled as 24 uniformly-distributed scatterers along its circumference, each with amplitude 0.35 to reflect its uniformly curved surface.
- Square was modeled with four corner scatterers (amplitude 1.0) and intermediate edge scatterers (amplitude 0.6).
- Rectangle was modeled identically to the Square but with asymmetric dimensions to encode aspect ratio.
- Triangle was modeled with three corner scatterers and interpolated edge scatterers.
- Oval was modeled with 24 elliptically-distributed scatterers with amplitudes varying around the perimeter.

Three size variants were generated for each shape: small (approximately 0.20 m primary dimension), medium (0.40 m), and large (0.60 m). All targets were placed within the radar's range envelope and span sizes representative of real-world objects.

3.3 Range-Angle Heatmap Generation

For each placement of a shape (defined by an orientation angle, a range, and a lateral offset from the 0° orientation), the simulation generated the IF signal received at each of the N_{ant} antennas.

The IF signal at antenna k for a single scatterer at coordinates (x, y) was computed as a sinusoid whose beat frequency encoded the range and whose inter-antenna phase progression encoded the angle. The contributions from all scatterers in a shape were superposed to produce the complete IF signal cube of dimensions $[N_s \times N_{\text{chirps}} \times N_{\text{ant}}]$, where $N_s = 160$ fast-time samples and $N_{\text{chirps}} = 1$ was sufficient for static-object simulation.

A 512-point Range FFT was applied across the fast-time dimension of each antenna's IF signal to produce a $[512 \times N_{\text{ant}}]$ complex-valued range profile. A 256-point Angle FFT was then applied across the antenna dimension to yield the final Range-Angle heatmap of dimensions $[512 \times 256]$. The magnitude of the result was retained for use as the CNN input. Figure 1 illustrates the canonical Range-Angle heatmaps for each shape class at 0° orientation and 4 m range with high signal-to-noise ratio.

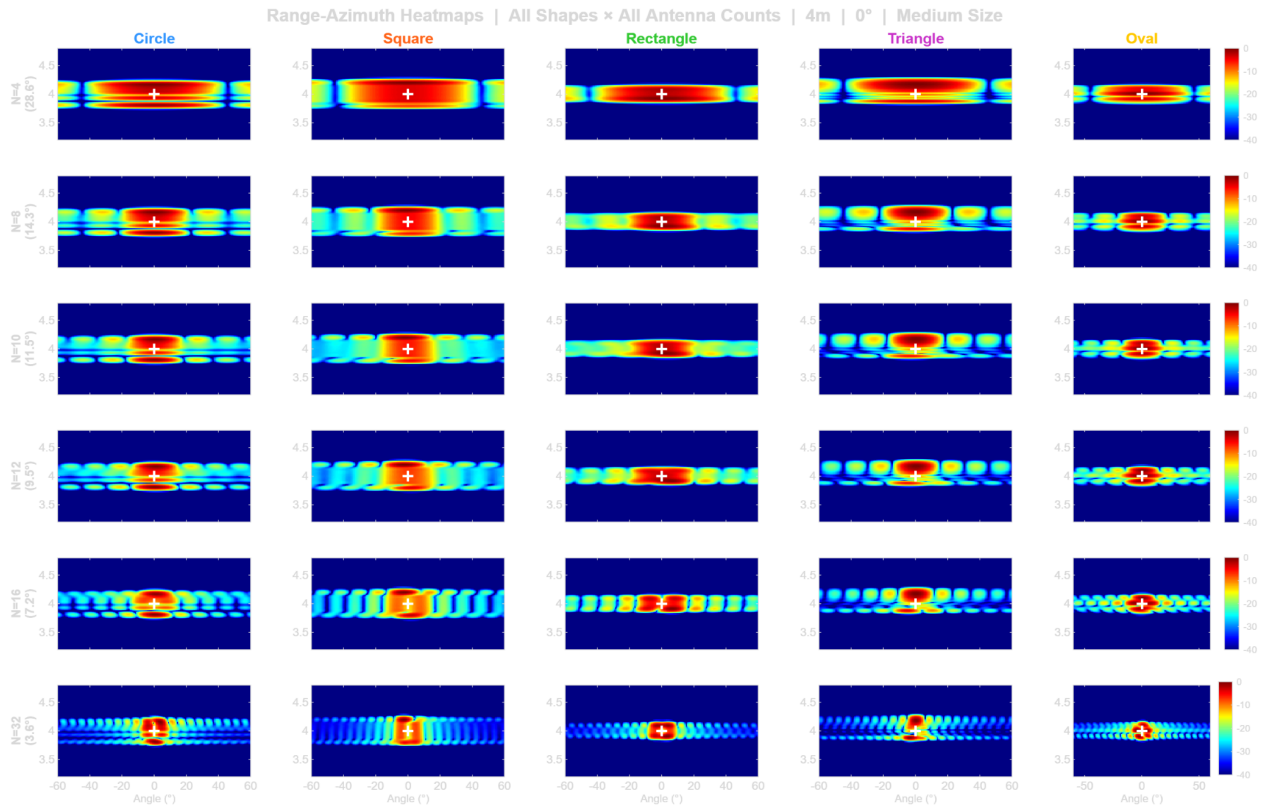


Figure 1

3.4 Domain Randomization

To narrow the anticipated sim-to-real gap, the simulation incorporated several domain randomization mechanisms. Thermal noise in the receiver was simulated by adding complex Gaussian noise to the signal, with the noise level adjusted to achieve four distinct SNR conditions: 5, 10, 20, and 30 dB. Free-space path loss was applied as an inverse-square attenuation of scatterer amplitudes with range to reproduce the physical fact that distant targets reflect less energy. Per-scatterer phase noise of standard deviation 0.05 radians was added to model oscillator imperfections. A $\pm 10\%$ amplitude jitter was applied independently to each scatterer of each sample to model material and surface irregularities. Finally, a small set of weak background scatterers was randomly placed in each frame to model environmental noise.

3.5 Dataset Generation and Splits

The full dataset was constructed by varying the placement of each shape across five axes: 36 orientations (0° to 350° in 10° steps), three sizes, four ranges (2, 3, 4, and 5 m), three lateral positions (-0.5 m, 0 m, +0.5 m), and four SNR levels (5, 10, 20, 30 dB). The Cartesian product yielded $36 \times 3 \times 4 \times 3 \times 4 = 5,184$ samples per shape, and $5,184 \times 5 = 25,920$ samples per antenna-count dataset. Six such datasets were generated, one for each value of $N_{\text{ant}} \in \{4, 8, 10, 12, 16, 32\}$, with the same geometric configurations across all six to ensure that observed accuracy differences could be attributed to angular resolution.

All four SNR variants of any given (shape, size, orientation, range, lateral) tuple were assigned to the same partition (training, validation, or test). This grouping ensured that the model could not memorize SNR-specific patterns of geometric configurations seen at other SNR levels in the

test set. The split ratio was 70% training, 15% validation, and 15% test, applied independently per shape class to maintain class balance.

3.6 CNN Architecture

The classifier architecture consisted of four sequential convolutional blocks followed by global average pooling and two fully-connected layers. The first block applied a 7×7 convolution with 16 output channels to capture broad structural patterns; the second block applied a 5×5 convolution with 32 channels for mid-level features; the third and fourth blocks each applied 3×3 convolutions with 64 and 128 channels respectively for fine-grained discrimination. Each block was followed by Batch Normalization, a ReLU activation, and 2×2 max-pooling. Global Average Pooling produced a 128-dimensional feature vector, which was passed through a 64-unit fully-connected layer with dropout (rate 0.4) and a final 5-unit linear layer to produce the shape class logits. The architecture was held constant across all six antenna-count experiments to ensure that observed differences in accuracy could be attributed to data. Table 2 summarizes the layer dimensions.

Table 2

Block	Layer	Output Shape
Input	Range-Angle Heatmap	$1 \times 512 \times 256$
Block 1	Conv 7×7 + BN + ReLU + MaxPool	$16 \times 256 \times 128$
Block 2	Conv 5×5 + BN + ReLU + MaxPool	$32 \times 128 \times 64$
Block 3	Conv 3×3 + BN + ReLU + MaxPool	$64 \times 64 \times 32$
Block 4	Conv 3×3 + BN + ReLU + MaxPool	$128 \times 32 \times 16$
GAP	Global Average Pooling	128
FC1	Linear + ReLU + Dropout(0.4)	64

Block	Layer	Output Shape
FC2	Linear	5

3.7 Training Procedure

Each model was trained using the Adam optimizer with an initial learning rate of 3×10^{-4} and standard cross-entropy loss. Training proceeded with mini-batches of size 128 for up to 60 epochs, with early stopping triggered when validation loss failed to improve for eight consecutive epochs. A ReduceLROnPlateau scheduler reduced the learning rate by a factor of 0.5 whenever validation loss plateaued for four epochs. Gradient clipping with maximum norm 1.0 was applied at every step to prevent training instability. To estimate variance, each (N_{ant}) experiment was repeated three times with random seeds $\{0, 1, 2\}$. Training was performed on a Google Colab A100 GPU.

3.8 Validation Pipeline

Two sim-to-real test methodologies were developed for validation and inference. The first used the MATLAB Radar Toolbox, which generates Range-Angle heatmaps using FMCW waveform models, real antenna beam patterns, thermal noise from radiometric equations, and phase noise models. The second methodology used shapes designed visually in Tinkercad (*a browser-based CAD tool*), exported as STL meshes, and converted to point-scatterer models. Both pipelines produced heatmaps in the same format as the training data, which allowed direct prediction by the trained model.

4. Results

4.1 Overall Classification Accuracy

Across the eighteen total training runs (six antenna counts $\times$ three random seeds), the mean test-set classification accuracy ranged from 94.3% at $N_{\text{ant}} = 4$ down to 64.6% at $N_{\text{ant}} = 16$. Table 3 summarizes the mean accuracy and standard deviation for each antenna count, and Figure 2 visualizes the same data with error bars.

Table 3

N_{ant}	Angular Resolution	Mean Accuracy	Std. Deviation	Best Seed
4	28.6°	94.3%	$\pm 0.3\%$	94.5%
8	14.3°	83.4%	$\pm 0.8\%$	84.1%
10	11.5°	81.4%	$\pm 1.6\%$	82.8%
12	9.6°	80.2%	$\pm 2.9\%$	82.6%
16	7.2°	64.6%	$\pm 29.4\%^*$	81.4%
32	3.6°	81.0%	$\pm 1.6\%$	82.7%

** $N = 16$ standard deviation is dominated by a single seed-0 training run that exhibited training divergence; see Section 5.2 for discussion.*

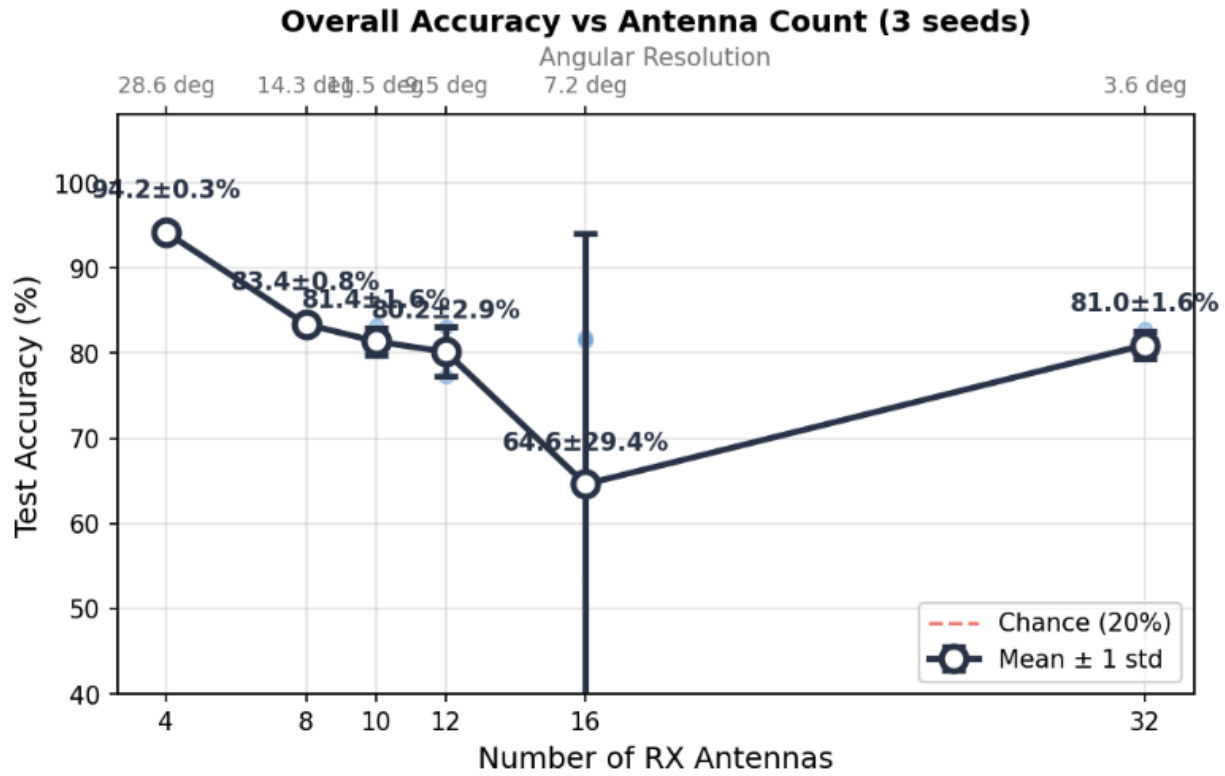


Figure 2

Two features of these results stand out. The four-antenna configuration produced the highest mean test accuracy of any configuration evaluated, contradicting the working hypothesis that accuracy would scale with antenna count. Second, the standard deviation at $N = 16$ is more than thirty times larger than at any other configuration, which indicates that the result at this configuration is dominated by a variation in the training-process rather than a problem from the data.

4.2 Per-Class Accuracy

Table 4 reports the per-class accuracy at three representative antenna counts, drawn from the best seed at each configuration. Figure 3 visualizes the same data as a grouped bar chart.

Table 4

Shape	N = 4	N = 8	N = 32
Circle	100.0%	100.0%	100.0%
Square	99.5%	99.0%	99.5%
Rectangle	80.9%	68.8%	47.4%
Triangle	96.4%	99.0%	82.1%
Oval	79.2%	83.4%	99.4%

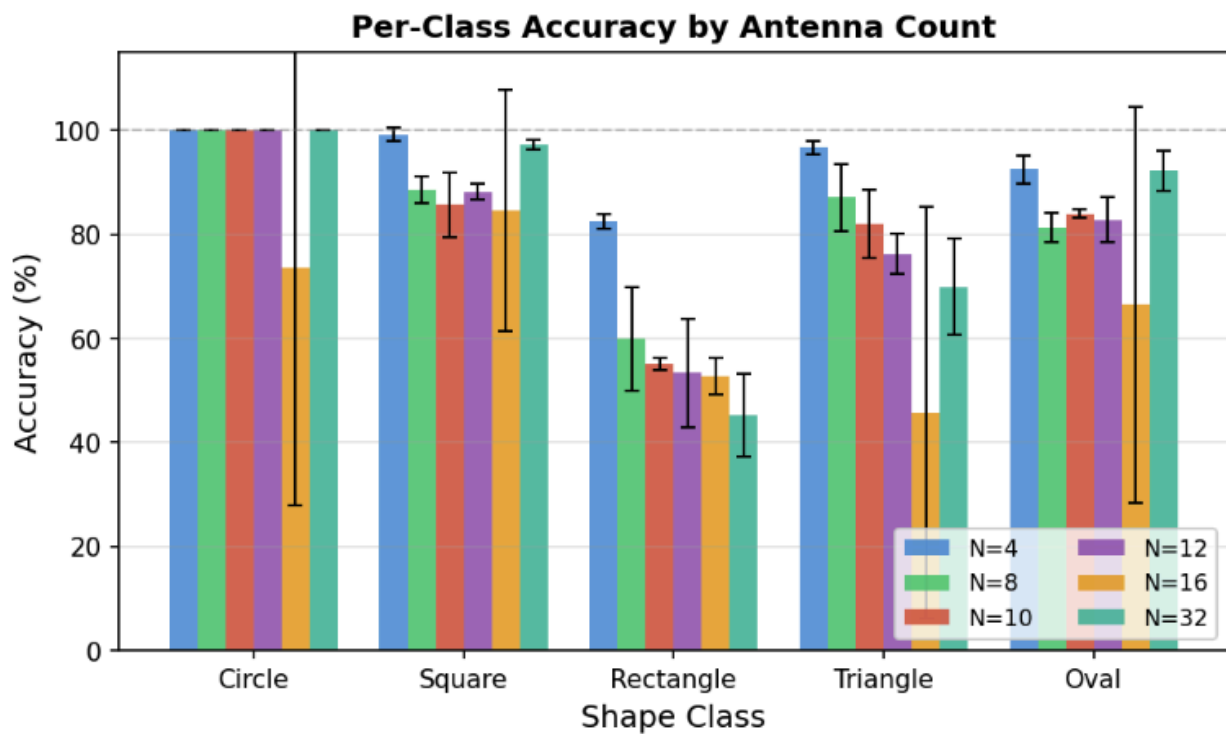


Figure 3

4.3 Confusion Matrices

Figure 4 presents the confusion matrix at $N = 4$, in which the diagonal is dominated by correct classifications. Figure 5 presents the confusion matrix at $N = 32$, where the rate of Rectangle samples being misclassified as Triangle has increased compared to $N = 4$.

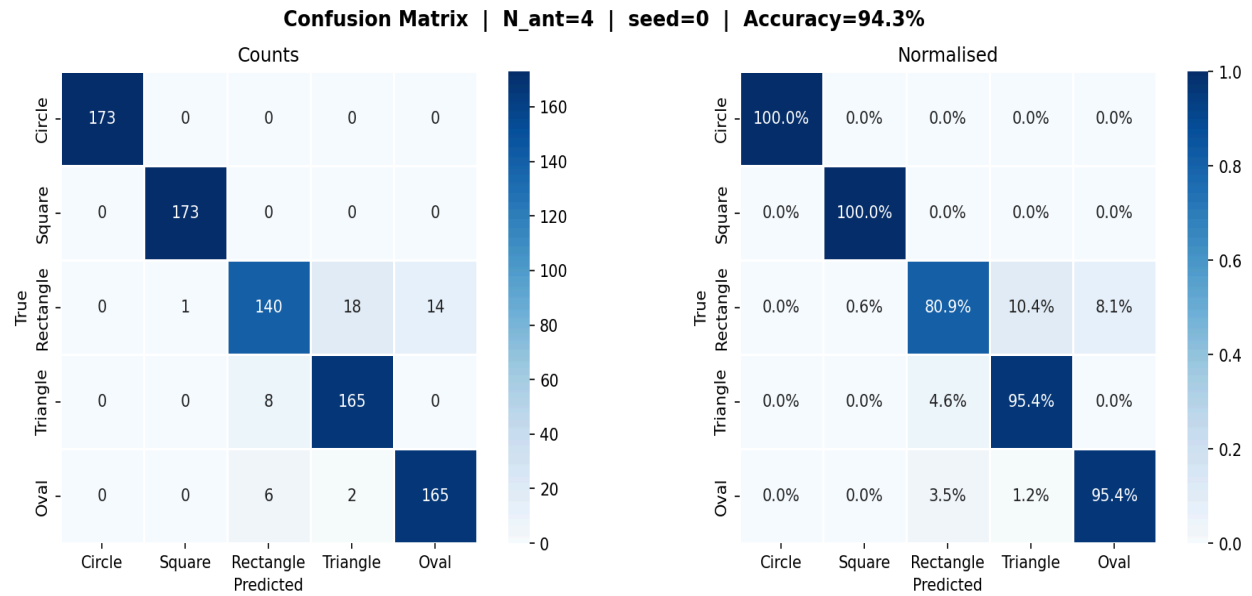


Figure 4

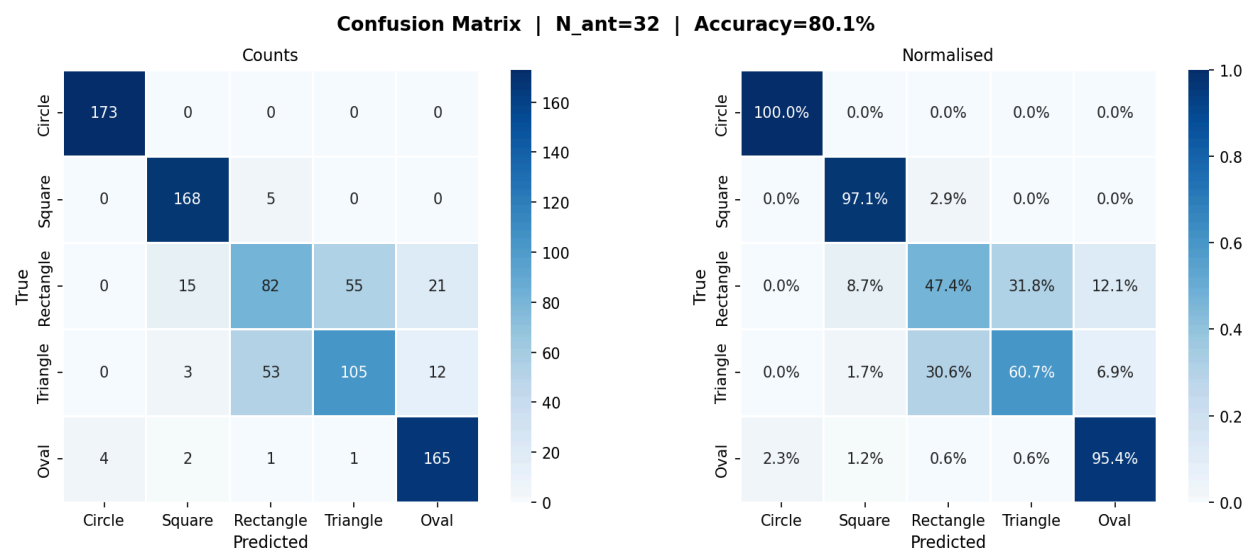


Figure 5

4.4 Training Dynamics

Training history curves illustrated qualitatively similar behavior across antenna counts. Figure 6 displays the training and validation loss and accuracy curves at $N = 4$, showing convergence.

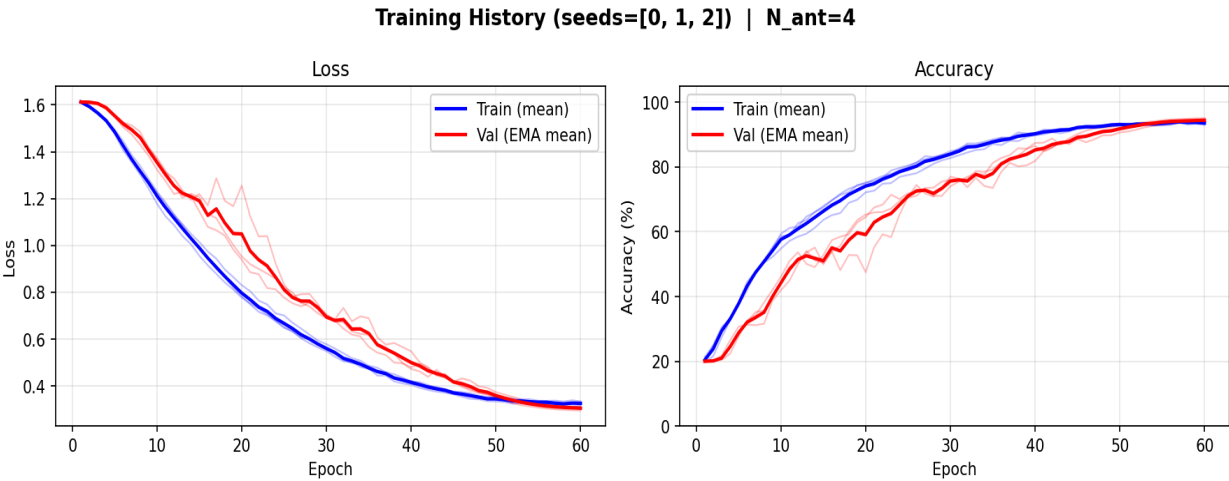


Figure 6

4.5 Validation and Inference

Both validation and inference were conducted by running held-out heatmaps generated via the MATLAB Radar Toolbox and the Tinkercad-STL pipeline through the trained model. Figure 7 shows an oval generated from a Tinkercad STL that was correctly classified at 86.6% confidence. A full quantitative model evaluation was not conducted in this work, as it requires real radar hardware data which remains a direction for future work.

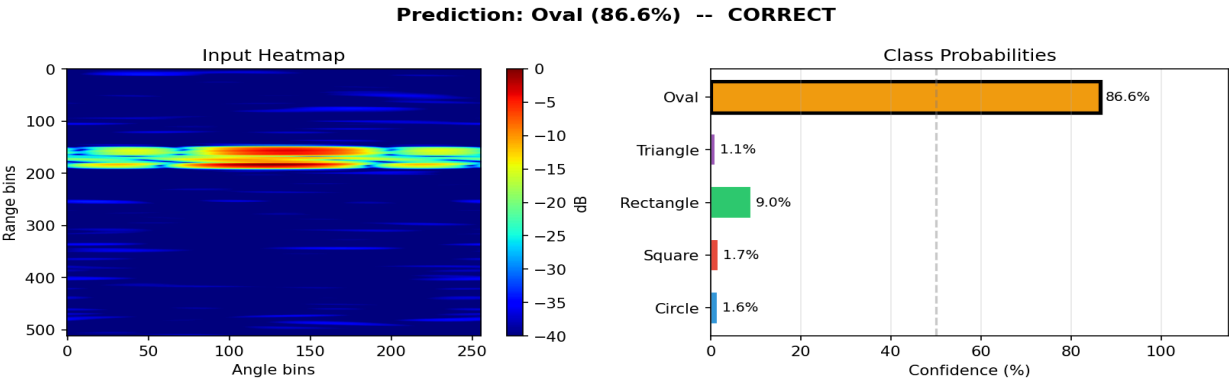


Figure 7

5. Discussion

5.1 Why Four Antennas Outperformed Higher Counts

The most consequential finding of this study was the non-linear relationship between antenna count and classification accuracy. The working hypothesis predicted that more antennas would improve accuracy by providing finer angular resolution, yet the data showed precisely the opposite pattern, with the four-antenna configuration achieving ~94% accuracy while configurations with eight or more antennas plateaued in the 80% range or below. The explanation lies in the per-class accuracy breakdown.

The Rectangle class, which served as the dominant accuracy bottleneck, shares its four-corner structure with the Square class, differing only in aspect ratio. The principal discriminating feature between Rectangle and Square is therefore the spread of scatterers in the range dimension (depth) versus the cross-range dimension (width). With only four antennas, the angular dimension is poorly resolved and the model is forced to rely on the range dimension to distinguish Rectangle from Square. With higher antenna counts the angular dimension becomes resolvable, and the model shifts its feature reliance from range to angle during training. This shift is harmful for shapes like Rectangle. At non-zero orientations, a Rectangle's angular signature closely resembles the three-corner pattern of a Triangle, so the angular features the model now attends to are precisely the ones that confuse the two classes. The data confirm this interpretation: at $N = 32$, Rectangle is most frequently misclassified as Triangle, not as Square. This interpretation also explains why the Oval class, which is rotationally smooth and lacks corner reflectors, benefits from higher antenna counts. The discriminating feature between Oval and Circle is the elongation, which is most cleanly resolved when the angular dimension carries fine-grained information about the cross-range extent of the target. The opposite trends for

Rectangle (degrading with N) and Oval (improving with N) provide internal consistency to the explanation: angular resolution helps when the discriminating feature is angular, and hurts when the discriminating feature is radial and cross-class confusion in the angular domain dominates.

5.2 The $N = 16$ Anomaly and Training Stability

The high standard deviation at $N = 16$ was driven by a single seed in which the model failed to converge. Validation loss diverged from epoch 10 onwards while training loss continued to decrease, and early stopping terminated the run at near-chance accuracy. The remaining two seeds at $N = 16$ converged successfully, which indicated that the failure was a stochastic training artifact rather than a property of the dataset. An evaluation protocol with five or more seeds would likely have yielded a smoother accuracy curve, but even with the failed seed excluded, the $N = 16$ mean would still fall below the $N = 4$ mean, preserving the central finding.

5.3 Implications for Hardware Design

These findings have direct implications for the design of FMCW radar systems intended for indoor object recognition. For shape classification, about four antennas appear to be sufficient, and additional antennas do not produce any meaningful improvements. This finding is important because consumer-grade FMCW radar chips such as the TI IWR1642 come with exactly four receive (RX) antennas, which indicates that off-the-shelf hardware is well-matched to this application.

5.4 Limitations

There were several limitations to this study. To begin with, all training and primary evaluation data were generated through simulation, and no real-hardware validation was performed.

Although the simulation incorporated physically motivated noise models and randomization, the absolute accuracy figures reported here are upper bounds on the accuracy achievable on real-hardware data. A second limitation is the size of the shape class which was restricted to only five shapes and therefore did not test the model on more complex shapes, which would have likely presented a more harder classification problem. A third limitation concerns the Rectangle class, which was generated at a single 2:1 aspect ratio. This may have made the Rectangle/Square boundary harder than necessary. A wider training distribution over aspect ratios could plausibly improve the Rectangle accuracy by a great amount.

6. Conclusion and Future Work

6.1 Summary of Findings

This study demonstrated that a Convolutional Neural Network operating on FMCW Range-Angle heatmaps could classify the geometric shape of a static target with mean test accuracy of ~94%, using a four-element receive antenna array consistent with consumer-grade radar hardware. An empirical comparison across six antenna counts (4, 8, 10, 12, 16, and 32) revealed that additional antennas did not improve classification accuracy. The Rectangle class proved the most difficult to classify, while the Circle class achieved near-perfect classification across all configurations. The data supported the interpretation that the discriminating feature between Rectangle and other shapes resides primarily in the range dimension rather than the angular dimension, and that increased angular resolution introduced additional sources of cross-class confusion.

Three contributions emerged from this work. First, a complete physics-based simulation and machine-learning pipeline for FMCW radar shape classification was developed and validated end-to-end. Second, a sim-to-real testing methodology was established, comprising both MATLAB Radar Toolbox-based and Tinkercad-STL-based heatmap generation for held-out evaluation. Third, an empirical study of antenna count's effect on classification accuracy yielded the finding that minimal antenna configurations are sufficient, and possibly preferable, for the shape classification task.

6.2 Future Work

Several directions follow from these findings. First and most importantly, real-hardware validation using the Texas Instruments IWR1642 evaluation board would transform the simulation-only results into a complete sim-to-real demonstration. A few real-hardware samples per shape class would suffice to fine-tune the trained model on real data. Also, expanding the Rectangle distribution in future model training to include multiple aspect ratios would directly address the bottleneck class identified in this study.

A natural next step is to extend the current 2D shape classification work to the recognition of real 3D objects. The infrastructure developed in this work transfers directly to that problem with two key additions: incorporating the Doppler dimension to capture motion signatures from objects in motion, and transitioning to a 3D Range-Doppler-Angle representation processed by a 3D CNN. The flat-shape classification results presented here provide a foundation for that follow-on work.

7. References

- [1] Ghasempour, Y. (2023). Lecture 32: Radar Systems. ECE/COS 368 Course Notes, Princeton University.
- [2] Texas Instruments. (2020). IWR1642 Single-Chip 76-GHz to 81-GHz mmWave Sensor. Technical Datasheet, SWRS212B.
- [3] MATLAB Radar Toolbox Documentation. (2024). The MathWorks, Inc., Natick, MA.

Appendix A. Per-Seed Experimental Results

Table A1 reports the test-set accuracy for every individual training run, organized by antenna count and random seed.

Table A1

N_{ant}	Seed 0	Seed 1	Seed 2	Mean $\pm$ Std
4	94.5	94.0	94.1	94.3 ± 0.3
8	84.1	82.6	83.5	83.4 ± 0.8
10	82.8	80.0	81.4	81.4 ± 1.6
12	82.6	76.8	81.2	80.2 ± 2.9
16	30.6*	81.4	81.0	64.6 ± 29.4
32	82.7	79.8	80.6	81.0 ± 1.6

Appendix B. Hyperparameter Specification

Table B1 lists the complete set of hyperparameters used during model training. These values were held constant across all six antenna-count experiments.

Table B1

Hyperparameter	Value
Optimizer	Adam ($\beta_1 = 0.9$, $\beta_2 = 0.999$)
Initial learning rate	3×10^{-4}
Loss function	Cross-entropy
Batch size	128
Maximum epochs	60
Early-stopping patience	8 epochs

Hyperparameter	Value
LR scheduler	ReduceLROnPlateau, factor 0.5, patience 4
Gradient clipping	Max norm 1.0
Dropout rate	0.4 (FC1 only)
Weight initialization	PyTorch default
Random seeds	{0, 1, 2}
Train / Val / Test split	70% / 15% / 15%

Appendix C. Code and Data Availability

All MATLAB simulation scripts, Python training scripts, results and trained models produced during this study are available in the following GitHub repository: [[Link](#)] .